Mohamed El Amraoui

AI Engineer · Data Scientist · Data Engineer · Software Engineer

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PROFILE

Master's graduate in Data Science & Intelligent Systems with a versatile profile spanning Generative AI (LLMs, RAG, AI agents), Machine Learning, data engineering, and full-stack software development. I have designed and shipped complete applications end-to-end: from data extraction and structuring to backend deployment and user-facing interfaces, including AI model integration. Curious, rigorous, and autonomous, I adapt quickly to different technical contexts and learn new technologies fast.

PROFESSIONAL EXPERIENCE

AI Developer Intern (Final-Year Internship) *Mar. 2026 – Jul. 2026*
Capgemini Engineering Casablanca

- Built a desktop application (Python, PySide6, FastAPI) automating the verification of technical assemblies, replacing a manual process.
- Automated data extraction from PDF documents using **Gemini** multimodal vision, with structured storage in **PostgreSQL**.
- Implemented a **RAG** system (LangChain, ChromaDB) enabling natural-language querying of technical standards, with source traceability.
- Designed an AI assistant using the **Agent Skills** pattern and automated report generation.
- Delivered a documented REST API and automated statistical computations with anomaly detection.

AI Developer Intern *Jul. – Sep. 2025*
Mosaic Mind Solutions Rabat

- Built an **MCP (Model Context Protocol)** routing agent within a multi-agent chatbot architecture.
- Integrated multiple LLM providers (OpenAI, Gemini, Groq, Stability AI) for NLP and image generation.
- Integrated AI agents into a React/TypeScript frontend and a Flutter mobile app, connected to Spring Boot APIs.
- Collaborated in an Agile environment with Git-based version control and continuous integration.

Full Stack Developer *Apr. – Jun. 2024*
CHU Hassan II Fès

- Designed a relational data model and built a web application (React, Spring Boot) for staff scheduling.
- Implemented secure role-based access control (IAM/RBAC) and conducted functional testing.

PERSONAL PROJECTS

AIMAAK – Darija Chatbot Assistant *Mar. – Jun. 2025*
Injaz Al-Maghrib Competition FST Fès

- Built a mobile Flutter application with an NLP-powered chatbot in Darija for small Moroccan businesses.

Diabetes Prediction with Machine Learning *Mar. 2025*
Academic Project FST Fès

- Implemented and compared models (KNN, SVM, Random Forest), handled class imbalance (SMOTE), and optimized hyperparameters.

Bank Customer Segmentation – K-Means *Feb. 2025*
FST Fès × University of Limerick Morocco/Ireland

- Performed unsupervised clustering and exploratory analysis to define customer segments and marketing recommendations.

TECHNICAL SKILLS

Generative AI & LLMs : Gemini, OpenAI, Groq, LangChain, LangGraph, RAG, Prompt Engineering, MCP, AI Agents

Machine Learning : Python, Scikit-learn, TensorFlow, PyTorch, NLP, Computer Vision, Deep Learning

Data Engineering : ETL pipelines, PostgreSQL, MySQL, MongoDB, Redis, ChromaDB, Spark/Hadoop (fundamentals)

Software Development : Java (Spring Boot), Python (FastAPI), React, TypeScript, Flutter, SQL
Cloud & DevOps : Docker, Kubernetes, Git/GitHub, CI/CD (fundamentals), Oracle Cloud Infrastructure (certified)
Security : Authentication, IAM/RBAC, encryption (AES, RSA)

EDUCATION

M.Sc. Data Science & Intelligent Systems *2024 – 2026*
FST Fès – Morocco
Machine Learning, Deep Learning, NLP, Big Data, Cloud Computing

B.Sc. Computer Science *2020 – 2024*
FST Fès – Morocco
Programming, Databases, Algorithms, Systems

CERTIFICATIONS

Oracle Cloud Infrastructure 2024 AI Foundations Associate – Oracle (Jan. 2025)
Extending AI with Agent Skills – Udemy (May 2026)
MCP Crash Course – Udemy (Jul. 2026)
LangChain and LangGraph Crash Course – Udemy (Jul. 2026)
Data Mining with Python Workshop – University of Limerick & FST Fès (Feb. 2025)
Git & GitHub – 365 Data Science (Nov. 2024)

LANGUAGES

French: Fluent **English:** Working proficiency **Arabic:** Native

Available for AI Engineer, Data Scientist, Data Engineer, or Software Engineer roles.